

Note 15.3 Area by Double Integration

Areas of Bounded Regions in the Plane

Definition

The **area** of a closed, bounded plane region R is

$$A = \iint_R dA$$

It can be obtained by take $f(x, y) = 1$ and the Riemann sum turns to the sum of areas.

Average Value

Definition

Average value of f over R :

$$\text{Ave} = \frac{1}{A} \iint_R f dA$$

Exercises

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